

Research Memo: Does the Market Correctly Value Novel Innovation?

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1 Part 1: Empirical Proxies

1.1 1A. Measuring Patent Novelty

Definition. I define patent novelty as the degree to which a patent’s technological content is dissimilar from all prior art. A patent is highly novel if its ideas have few precedents in the existing patent landscape; it is incremental if it closely resembles previously patented technologies.

Measure. I adopt the backward similarity measure from Kelly, Papanikolaou, Seru, and Taddy (2021, *AER: Insights*). Their approach converts each patent’s full text (abstract, claims, and detailed description) into a TF-IDF vector representation, then computes the cosine similarity between a given patent and the corpus of all previously granted patents. The resulting *backward similarity* score ranges from 0 to 1, where higher values indicate greater resemblance to existing technologies.

I construct the novelty variable as:

$$Novelty_i = 1 - BackwardSimilarity_i \tag{1}$$

so that higher values correspond to greater novelty. Specifically, I use their 5-year backward similarity measure (bsim5), which compares each patent to those granted in the preceding five years, providing a more temporally relevant benchmark.

Why this measure. Text-based similarity has several advantages over citation-based alternatives. First, it captures technological content directly rather than relying on citation networks, which are influenced by examiner behavior, strategic citing, and truncation bias (Hall, Jaffe, and Trajtenberg 2001). Second, the Kelly et al. measure covers all U.S. patents

from 1836 to 2022 with a consistent methodology, providing a complete and comparable dataset. Third, it has been validated against both expert assessments and forward-looking outcomes in a top peer-reviewed journal.

Assumptions. The measure relies on three key assumptions: (1) patent text accurately reflects the underlying technology, rather than merely reflecting drafting conventions of patent attorneys; (2) TF-IDF cosine similarity adequately captures technological distance—i.e., textually dissimilar patents are technologically dissimilar; and (3) the set of all prior patents is the appropriate reference corpus for gauging novelty.

Limitations. First, text similarity is an imperfect proxy for technological similarity. Two patents may use different terminology to describe similar inventions (synonymy), or similar language to describe different technologies (polysemy). Second, patent attorneys may strategically draft claims using broad, generic language to maximize protection scope, artificially lowering textual similarity without reflecting genuine technological novelty. Third, the TF-IDF bag-of-words model ignores word order and semantic relationships, potentially missing conceptual similarities that more sophisticated NLP methods (e.g., transformer-based embeddings) could capture.

Optional Extension: Novelty vs. Quality. Novelty and quality are conceptually distinct. A patent can be highly novel yet commercially worthless (e.g., solving a problem no one has), or incremental yet extremely valuable (e.g., a critical process improvement in semiconductor manufacturing). I propose measuring quality separately using Kelly et al.’s *forward similarity* (fsim5), which captures how much a patent influences subsequent technological development. The distinction matters for our research question because if we observe that novel patents receive lower market valuations, we must rule out the possibility that this reflects genuinely lower quality rather than market mispricing. Including a quality control variable in our regression helps disentangle these two channels.

An important caveat: forward similarity for 2021–2022 patents suffers from truncation bias, as only a few years of subsequent patents are available to compute forward-looking measures. Kelly et al. note that their importance and breakthrough indicators are reliable only up to 2016 for the 5-year forward measure. For our sample period, forward similarity should therefore be interpreted with caution and treated as a noisy proxy for quality.

1.2 1B. Measuring Technology Uncertainty

Definition. Technology uncertainty refers to the degree of difficulty investors face in assessing the economic value of a patent at the time of its grant. When investors cannot rely on precedents or familiar benchmarks to evaluate a new patent, uncertainty is high.

I propose two approaches to measure this construct:

Approach 1: Patent Value Dispersion within Technology Classes.

Conceptual justification. If the market values of patents within a given technology area vary enormously—some worth billions, others worth almost nothing—then investors facing a new patent in that area have little basis for forming a precise valuation. The high dispersion of realized values serves as a revealed measure of valuation difficulty. This logic aligns with Pastor and Veronesi (2003, *JPE*), who model technology uncertainty as the unpredictability of a new technology’s future profitability, and with Diether, Malloy, and Scherbina (2002, *JF*), who use analyst forecast dispersion as a proxy for information uncertainty.

Implementation. For each CPC subclass $\times$ grant year cell, I compute the coefficient of variation (CV) of KPSS patent values (xi_real):

$$Uncertainty_{c,t} = \frac{\sigma(xi_real)_{c,t}}{\mu(xi_real)_{c,t}} \quad (2)$$

where c indexes CPC subclasses and t indexes grant years. I require a minimum of 10 patents per cell to ensure the statistic is meaningful. The CV is scale-free, allowing meaningful comparison across technology areas with different average patent values.

Data requirements. Only KPSS patent values and CPC classification data, both of which are readily available.

Limitations. First, high value dispersion may reflect genuine heterogeneity in patent quality within a technology class rather than information uncertainty per se. Second, this measure is constructed from the distribution of the dependent variable (KPSS values), which could introduce mechanical correlation. While the measure is aggregated at the CPC subclass level and the dependent variable is at the patent level, this concern deserves acknowledgment. Third, the measure captures cross-sectional dispersion, which may not perfectly correspond to ex ante uncertainty at the time of patent grant.

Approach 2: Examination Duration Volatility.

Conceptual justification. The time between patent filing and grant reflects how difficult it is to evaluate the patent’s claims. If examination durations within a technology class vary widely, it suggests that even expert USPTO examiners face inconsistent difficulty in assessing patents in that area—a signal that investors likely face similar evaluative challenges.

Implementation. Compute the standard deviation of examination duration (grant date minus filing date, in months) within each CPC subclass $\times$ year cell.

Data requirements. Filing dates and grant dates, available in the KPSS dataset.

Limitations. Examination duration is influenced by many non-technical factors: examiner workload, applicant response times, Track One prioritized examination requests, and

continuation filings. These administrative sources of variation may dominate any signal about technological uncertainty. Moreover, examiner uncertainty and investor uncertainty are fundamentally different constructs.

Choice of implementation. I implement Approach 1 (value dispersion) because it directly captures the unpredictability of economic outcomes that investors face, which is conceptually closest to the information uncertainty mechanism in our hypothesis. Approach 2, while offering the advantage of using an independent data source, is too contaminated by administrative noise to serve as a reliable uncertainty proxy.

Ideal alternative. If data were available, the conceptually cleanest measure would be analyst earnings forecast dispersion (from I/B/E/S) for the patent’s assignee firm around the grant date. This directly captures disagreement among professional evaluators and has been validated extensively in the behavioral finance literature. However, this measure is only available for publicly traded firms with analyst coverage and operates at the firm rather than patent level.

1.3 1C. Measuring the Market’s Valuation of Patents

As specified in the assignment, I use the patent-level economic value estimates from Kogan, Papanikolaou, Seru, and Stoffman (2017, *QJE*). Their measure (*xi_real*) combines the stock market reaction around patent grant dates with a structural model to estimate each patent’s private economic value in real (1982) dollars. The data is publicly available and covers patents through 2024.

2 Part 2: Research Design

2.1 Main Specification

To test whether investors undervalue novel patents, I estimate:

$$\ln(xi_real_i) = \beta_1 Novelty_i + \beta_2 NumClaims_i + \beta_3 \ln(BackwardCites_i + 1) + \delta_t + \delta_c + \varepsilon_i \quad (3)$$

where:

- xi_real_i is the KPSS patent market value for patent i
- $Novelty_i = 1 - BackwardSimilarity_i$ from Kelly et al. (2021)
- $NumClaims_i$ controls for patent scope (more claims typically indicate broader protection)
- $BackwardCites_i$ controls for the breadth of the patent’s knowledge base

- δ_t denotes grant year fixed effects (2021 vs. 2022)
- δ_c denotes CPC section fixed effects (A through H and Y)
- Standard errors are clustered at the CPC subclass level

The dependent variable is log-transformed for two reasons. First, patent values are heavily right-skewed, with a small number of patents accounting for a disproportionate share of total value; the log transformation produces a more symmetric distribution suitable for OLS. Second, log-transforming the dependent variable means that coefficients are interpreted as approximate percentage changes—a one-unit increase in novelty is associated with a $\beta_1 \times 100\%$ change in patent value. This is economically more meaningful than absolute dollar differences, since a 10% valuation discount is equally significant for a \$1 million patent and a \$100 million patent.

The hypothesis predicts $\beta_1 < 0$: controlling for other patent characteristics, more novel patents receive lower market valuations, consistent with systematic undervaluation.

2.2 Mechanism Test

To test whether information uncertainty is the channel through which novelty affects valuation, I estimate:

$$\ln(xi_real_i) = \beta_1 Novelty_i + \beta_2 Novelty_i \times Uncertainty_{c,t} + \beta_3 Uncertainty_{c,t} + \gamma X_i + \delta_t + \delta_c + \varepsilon_i \quad (4)$$

where $Uncertainty_{c,t}$ is the coefficient of variation of KPSS values within CPC subclass c and year t , and X_i includes the same control variables.

If information uncertainty mediates the novelty discount, we expect $\beta_2 < 0$: the negative relationship between novelty and market value should be amplified in technology areas where valuation uncertainty is higher.

Additionally, a necessary condition for the mechanism is that novelty is positively associated with uncertainty. This can be verified with a “first stage” regression:

$$Uncertainty_{c,t} = \alpha_1 \overline{Novelty}_{c,t} + controls + \varepsilon_{c,t} \quad (5)$$

where $\overline{Novelty}_{c,t}$ is the average novelty within CPC subclass c and year t . If $\alpha_1 > 0$, it would confirm that more novel technology areas indeed exhibit greater valuation uncertainty. Without this link established, the mechanism test loses its conceptual foundation—even a significant interaction term would be difficult to interpret if novelty and uncertainty are unrelated.

2.3 Control Variables and Fixed Effects

Control variables:

- *Number of claims* controls for the breadth of patent protection. Patents with more claims tend to cover a wider technological space and are typically more valuable.
- *Log backward citations* controls for the size of the patent’s prior art base. More citations may indicate either a well-researched invention or a crowded technological area.

Fixed effects:

- *Year fixed effects* absorb macroeconomic conditions, market-wide sentiment shifts, and policy changes between 2021 and 2022 (e.g., post-COVID technology sector dynamics).
- *CPC section fixed effects* absorb systematic differences in patent valuation across broad technology domains (e.g., pharmaceutical patents are valued differently from software patents due to different market structures and regulatory environments).

Clustering. Standard errors are clustered at the CPC subclass level (~ 670 clusters) because patents within the same narrow technology field likely share common valuation shocks (same set of potential investors, similar information environment, correlated technological trajectories).

A note on fixed effects granularity. One might consider using CPC subclass fixed effects rather than section fixed effects for more stringent within-technology comparisons. However, since the uncertainty measure is constructed at the CPC subclass $\times$ year level, including subclass fixed effects would absorb most of the variation in uncertainty, making it impossible to estimate the mechanism test. The choice of section-level fixed effects therefore reflects a deliberate trade-off: it controls for broad cross-domain differences while preserving sufficient within-section variation to identify the effects of both novelty and uncertainty. As a robustness check, one could use CPC group-level fixed effects (intermediate granularity) to test sensitivity to this choice.

2.4 Results

Table 1: Does the Market Discount Novel Patents?

	(1) Main	(2) Mechanism
Novelty	0.0771 (0.0739)	0.0329 (0.2612)
Tech Uncertainty (CV)		-0.7505*** (0.2022)
Novelty x Uncertainty		0.0116 (0.1279)
Num Claims	0.0869*** (0.0058)	0.0834*** (0.0054)
ln(Backward Cites + 1)	0.3459*** (0.0253)	0.3350*** (0.0254)
Constant	0.4637** (0.1850)	1.6637*** (0.3951)
R-squared	0.1010	0.1308
Adj R-squared	0.1009	0.1308
N	194,937	194,937
Year FE	Yes	Yes
CPC Section FE	Yes	Yes
Cluster SE	CPC subclass	CPC subclass

Standard errors clustered at CPC subclass level in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 2: First Stage: Does Novelty Predict Technology Uncertainty?

	Dep Var: Tech Uncertainty (CV)
Mean Novelty	-0.3937*** (0.0637)
R-squared	0.088
N	680
Year FE	Yes

Unit of observation: CPC subclass $\times$ year.

* p<0.10, ** p<0.05, *** p<0.01

2.5 Critical Assessment

Result 1: Novelty does not predict lower patent values ($\beta_1 = 0.077$, $p = 0.30$). The coefficient is positive but statistically insignificant, meaning there is no evidence that the market systematically discounts novel patents. If anything, the positive point estimate suggests markets may slightly *reward* novelty in this sample. This does not support the central hypothesis.

Result 2: Technology uncertainty significantly reduces patent values ($\beta_3 = -0.75$, $p < 0.001$). This is the strongest finding: patents in high-uncertainty technology areas receive substantially lower market valuations. A one-standard-deviation increase in the CV of patent values within a CPC subclass is associated with approximately 75% lower patent value. This confirms that information uncertainty is a real friction in patent valuation, consistent with the theoretical framework of Pastor and Veronesi (2003).

Result 3: The interaction Novelty $\times$ Uncertainty is zero ($\beta_2 = 0.012$, $p = 0.93$). Even if uncertainty depresses patent values, this effect does not disproportionately fall on novel patents. The discount from uncertainty is equally severe for novel and incremental patents alike.

Result 4: The first stage contradicts the mechanism ($\alpha_1 = -0.39$, $p < 0.001$). This is the most important finding for evaluating the proposed mechanism. The hypothesis assumes a causal chain: novelty $\rightarrow$ higher uncertainty $\rightarrow$ lower valuation. But the first-stage regression reveals the opposite: technology areas with higher average novelty actually exhibit *lower* value dispersion. More novel CPC subclasses are more predictable, not less.

This negative first-stage result undermines the proposed mechanism at its foundation. Several explanations are possible:

- (a) *Mechanical artifact of the CV measure.* If novel technology areas tend to have uniformly low patent values (i.e., both the mean and standard deviation are low), the CV could be small simply because there is less absolute variation—not because investors face less uncertainty. In mature technology areas, the coexistence of high-value blockbuster patents and low-value marginal patents generates high dispersion. This suggests the CV may capture outcome heterogeneity rather than ex ante investor uncertainty.
- (b) *Novel technology areas may be more homogeneous.* If a CPC subclass is dominated by genuinely new technology, the patents within it may be more similar to each other (all exploring the same frontier), producing lower value dispersion. By contrast, mature technology areas may contain a wider mix of patent types.
- (c) *The uncertainty measure may not capture investor uncertainty.* The CV of realized KPSS values reflects ex post outcome heterogeneity, which is not the same as ex ante investor uncertainty. Mature fields with well-known but varied applications (e.g., phar-

maceutical formulations) may show high value dispersion even though investors understand the technology well.

- (d) *Text-based novelty may not align with market-perceived novelty.* Kelly et al.’s backward similarity measures textual distance from prior patents, but investors may perceive novelty through different channels—patent family structure, inventor track record, assignee reputation, or technology classification patterns. If text novelty and market-perceived novelty diverge, the first-stage correlation would be uninformative about the true mechanism.

Interpreting the full picture. Taken together, these results tell a coherent story: technology uncertainty matters for patent valuation, but it does not operate through the novelty channel as hypothesized. The market does not appear to discount novel patents, and novel technology areas are not characterized by higher uncertainty. The original hypothesis—that novelty $\rightarrow$ uncertainty $\rightarrow$ undervaluation—is not supported in this sample.

This null result is itself informative. It suggests either that (1) markets are reasonably efficient at incorporating novelty information into patent valuations, at least during 2021–2022; (2) the text-based novelty measure does not capture the dimension of novelty that creates valuation difficulty for investors; or (3) the 2021–2022 period is atypical due to post-COVID conditions that favored novel technologies.

Sample period limitation. The restriction to 2021–2022 is a major caveat. Two years provide limited statistical power and no ability to examine time-series variation. This period is characterized by unusual macroeconomic conditions—low interest rates, high growth expectations, pandemic-driven shifts in technology demand (mRNA, remote work technology, AI)—that may have temporarily reversed the typical novelty discount. A longer sample spanning different market regimes would be essential to draw definitive conclusions.

Alternative explanations.

1. *Quality confound.* Novel patents may genuinely be less valuable on average—not because markets misprice them, but because breakthrough ideas often fail commercially. To address this, one could control for forward citations or Kelly et al.’s forward similarity as a quality proxy.
2. *Selection effects.* Firms that produce novel patents may differ systematically from those producing incremental ones. For example, smaller firms with lower market capitalization may invest more in exploratory research, and the KPSS methodology may assign lower values to their patents due to lower stock price responsiveness. Firm-level controls or assignee fixed effects could help address this concern.
3. *Reverse causality.* Firms that are systematically undervalued by the market may invest more in novel research precisely because they face lower opportunity costs for risky

exploration. This selection channel would generate the same empirical pattern—a correlation between novelty and low valuation—without implying market mispricing of individual patents. Addressing this would require an instrumental variable strategy, such as exploiting quasi-random variation in patent grant timing or examiner leniency.

4. *Investor attention.* An alternative to the uncertainty channel is that investors simply pay less attention to novel patents because they fall outside familiar categories. Under this story, the mechanism is not uncertainty but neglect—and the remedy would not be better disclosure but greater analyst coverage. Distinguishing between uncertainty and inattention requires separate proxies for each (e.g., analyst coverage for attention, forecast dispersion for uncertainty).

Measurement limitations.

1. *Novelty:* The text-based measure is susceptible to patent drafting strategies. Attorneys at sophisticated firms may systematically craft language that appears novel or familiar, irrespective of the underlying technology.
2. *Uncertainty:* Constructing uncertainty from the dispersion of the dependent variable (KPSS values) within technology classes risks introducing mechanical correlation. While aggregation at the CPC subclass level mitigates this concern, it does not eliminate it entirely. Moreover, the negative first-stage result suggests that this measure may not adequately capture the type of uncertainty that investors face when evaluating novel patents. An ideal measure would use an independent information source such as analyst forecast dispersion from I/B/E/S.
3. *KPSS values:* The sample is restricted to patents assigned to publicly traded firms, excluding patents from universities, government labs, and private companies. This selection may bias findings if the novelty-valuation relationship differs for non-public entities.

Possible improvement. With additional time and data, I would make three changes. First, extend the sample to 2010–2022 to test whether the null result on novelty is specific to the post-COVID period. Second, replace the value-dispersion uncertainty measure with analyst forecast dispersion from I/B/E/S, which is conceptually independent of the dependent variable and directly captures investor disagreement. Third, incorporate patent disclosure specificity as a moderating variable, using the LLM-based scoring approach described in Part 3E.

3 Part 3E: Disclosure Specificity Measure

3.1 Definition

Patent disclosure specificity is not merely a matter of surface-level linguistic complexity (longer words or more numbers). Rather, I define it as the degree to which a patent disclosure enables a reader to concretely understand:

1. What technical problem the invention solves
2. What the technical solution is
3. Whether key technical parameters, structures, steps, or conditions are explicitly stated
4. Whether a reader can clearly distinguish the invention from prior art

In short, specificity is about whether the information conveyed is **concrete, identifiable, comprehensible, and actionable**—not simply whether the language is complex.

3.2 Why Shallow Text Features Are Insufficient

Traditional NLP approaches to measuring text specificity—such as numeric density, average word length, or type-token ratio—capture surface-level properties but miss the semantic core of what makes a disclosure specific. A patent abstract may contain long technical words yet still be vague about the actual invention mechanism. Conversely, a short abstract with simple language may precisely convey the key innovation. Specificity is fundamentally a semantic judgment that requires understanding the content, not just counting words.

3.3 Proposed Approach: LLM-Based Scoring with a Structured Rubric

I propose measuring disclosure specificity by using a large language model (LLM) as a scalable annotator, guided by a transparent, pre-defined rubric. The LLM’s role is not to invent its own standard, but to simulate a trained research assistant applying a consistent coding scheme at scale.

Scoring Rubric (1–5 scale):

- **1 (Very vague):** Description is entirely generic—uses phrases like “improve performance” or “enhance efficiency” with no concrete technical content. A reader cannot determine what the invention actually does.
- **2 (Mostly vague):** Mentions some technical objects or domains, but remains fundamentally unclear about the mechanism, method, or key parameters.

- **3 (Moderately specific):** Conveys the core technical content, but with limited detail. The technical pathway or implementation conditions are not fully specified.
- **4 (Specific):** Clearly describes the technical solution, key components/steps/parameters. A reader can form a reasonably clear picture of the invention.
- **5 (Highly specific):** Discloses the technical mechanism, structure, quantitative parameters, and implementation conditions in detail. Clearly distinguishes the invention from existing technology.

Implementation:

- *Input:* Patent abstract (or, where available, abstract + first independent claim)
- *Prompt design:* The LLM is instructed to first provide a brief justification (1–2 sentences), then assign a score from 1–5 according to the rubric. Requiring justification before scoring reduces arbitrary ratings and improves consistency.
- *Model choice:* A capable LLM such as Claude or DeepSeek, selected for cost-efficiency at scale.
- *Sample:* For a full implementation, one would score all $\sim 196,000$ patents in the 2021–2022 sample. For a pilot, a random subsample of 2,000–3,000 patents would suffice to establish distributional properties and validate the measure.

3.4 Validation Strategy

To ensure the LLM measure is not arbitrary, I propose a human validation protocol:

1. Randomly sample 200–300 patents from the scored set
2. Have one or more human annotators independently score them using the same 1–5 rubric
3. Compute inter-rater agreement between LLM and human scores using:
 - Pearson correlation (linear agreement)
 - Spearman rank correlation (ordinal agreement)
 - Cohen’s kappa (categorical agreement)
 - Bin-level concordance rates (e.g., proportion of scores within ± 1 point)

If agreement is high (e.g., Pearson $r > 0.6$, kappa > 0.4), this demonstrates that the LLM measure approximates human judgment under a transparent, replicable scoring rule—a standard that most shallow text features cannot meet.

3.5 Why LLM Scoring Outperforms Traditional NLP for This Task

Specificity is a holistic semantic judgment. It requires understanding whether a description communicates *what the invention is and how it works*, not just whether it uses long words or numbers. LLMs excel at precisely this type of judgment because they can:

- Parse the meaning of technical descriptions, not just their surface features
- Distinguish between genuinely informative content and boilerplate language
- Apply a consistent rubric across heterogeneous technology domains
- Handle the enormous stylistic variation across patent attorneys and technology fields

Traditional classifiers or dictionary-based approaches would require extensive feature engineering and domain-specific tuning to approximate this level of semantic understanding.

3.6 Limitations

- LLM scores may exhibit systematic biases (e.g., favoring certain writing styles or technology domains). Domain-specific calibration or within-CPC normalization could mitigate this.
- Reproducibility depends on model version and temperature settings; documenting exact model parameters is essential.
- The rubric, while structured, still involves subjective judgment—reasonable annotators (human or LLM) may disagree on borderline cases.
- Cost and computational constraints may limit full-sample scoring, though API costs for models like DeepSeek are modest (approximately \$2–5 for 200,000 abstracts).
- Due to time constraints in this project, I describe the methodology without implementing the full scoring pipeline. Implementation would be straightforward given access to the patent abstract data and an LLM API.

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